16/07/26 Exp no: 1 Network commands

Aim:

Study of various network commands used in Linux and window

Basic Network commands:

i) ARP - Displays and manages the app table, map IP addres

ii) hostname - Displays / changes the name of local computer.

```
C:\Users\shreya>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Users\shreya>hostname
ANGELINSHREYA
```

iii) ip config /all - Displays the IP address subnet mark, and default gateway

```
C:\Users\shreya>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : ANGELINSHREYA
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-13
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::d965:a9b0:8284:42fb%19(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 755630119
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-BF-2B-53-BC-FC-E7-E1-01-8A
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 5A-02-05-D9-C4-92
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 5E-02-05-D9-C4-92
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:
```

iv) nbtstat -a Displays and Co-net BIOS over TCP/IP and net BIOS name table

```
C:\Users\shreya>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

v) net - stat Display active network connections routing tables and listing ports

```
C:\Users\shreya>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:1042         ANGELINSHREYA 55112    ESTABLISHED
  TCP    127.0.0.1:1042         ANGELINSHREYA 55121    ESTABLISHED
  TCP    127.0.0.1:9012         ANGELINSHREYA 55114    ESTABLISHED
  TCP    127.0.0.1:13030        ANGELINSHREYA 49680    ESTABLISHED
  TCP    127.0.0.1:13031        ANGELINSHREYA 49714    ESTABLISHED
  TCP    127.0.0.1:49680        ANGELINSHREYA 13030    ESTABLISHED
  TCP    127.0.0.1:49714        ANGELINSHREYA 13031    ESTABLISHED
  TCP    127.0.0.1:49744        ANGELINSHREYA 49745    ESTABLISHED
  TCP    127.0.0.1:49745        ANGELINSHREYA 49744    ESTABLISHED
  TCP    127.0.0.1:49752        ANGELINSHREYA 49753    ESTABLISHED
  TCP    127.0.0.1:49753        ANGELINSHREYA 49752    ESTABLISHED
  TCP    127.0.0.1:49758        ANGELINSHREYA 49759    ESTABLISHED
  TCP    127.0.0.1:49759        ANGELINSHREYA 49758    ESTABLISHED
  TCP    127.0.0.1:50100        ANGELINSHREYA 55106    ESTABLISHED
  TCP    127.0.0.1:55106        ANGELINSHREYA 50100    ESTABLISHED
  TCP    127.0.0.1:55112        ANGELINSHREYA 1042     ESTABLISHED
  TCP    127.0.0.1:55114        ANGELINSHREYA 9012     ESTABLISHED
  TCP    127.0.0.1:55118        ANGELINSHREYA 55119    ESTABLISHED
  TCP    127.0.0.1:55119        ANGELINSHREYA 55118    ESTABLISHED
  TCP    127.0.0.1:55121        ANGELINSHREYA 1042     ESTABLISHED
  TCP    127.0.0.1:58142        ANGELINSHREYA 58143    ESTABLISHED
  TCP    127.0.0.1:58143        ANGELINSHREYA 58142    ESTABLISHED
  TCP    127.0.0.1:58144        ANGELINSHREYA 58145    ESTABLISHED
  TCP    127.0.0.1:58145        ANGELINSHREYA 58144    ESTABLISHED
  TCP    127.0.0.1:60048        ANGELINSHREYA: 60049   ESTABLISHED
```

vi) path ping - This command traces route to the distination address then launch a 25 second test

```
PS C:\Users\shreya> pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

for each router along the way.

vi) ns look up - Queries DNS servers to find the ns domain name of a host

```
C:\Users\shreya>ns
DNS request timed out.
    timeout was 2 seconds.
Default Server:  UnKnown
Address:  2401:4900:50:9::280
```

viii) ping - It is the best way to test connectivity between two nodes.

```
PS C:\Users\shreya> ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t                  Ping the specified host until stopped.
                        To see statistics and continue - type Control-Break;
    -a                  To stop - type Control-C.
    -n count            Resolve addresses to hostnames.
    -l size             Number of echo requests to send.
    -f                  Send buffer size.
    -i TTL              Set Don't Fragment flag in packet (IPv4-only).
    -v TOS              Time To Live.
                        Type Of Service (IPv4-only. This setting has been deprecated
                        and has no effect on the type of service field in the IP
                        Header).
    -r count            Record route for count hops (IPv4-only).
    -s count            Timestamp for count hops (IPv4-only).
    -j host-list        Loose source route along host-list (IPv4-only).
    -k host-list        Strict source route along host-list (IPv4-only).
    -w timeout          Timeout in milliseconds to wait for each reply.
    -R                  Use routing header to test reverse route also (IPv6-only).
                        Per RFC 5095 the use of this routing header has been
                        deprecated. Some systems may drop echo requests if
                        this header is used.
    -S srcaddr          Source address to use.
    -c compartment      Routing compartment identifier.
    -p                  Ping a Hyper-V Network Virtualization provider address.
    -4                  Force using IPv4.
    -6                  Force using IPv6.
```

vii) path ping - This command traces route to the destination address then launches a 25 second test for each router along the way.

xi) Route - Route command is used to show / manipulate the IP routing table.

```
PS C:\Users\shreya> Route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*          .... Only prints those matching 157*
```

Result:

Thus the network commands are executed successfully.